Electroweak Scale and Fermion Masses from Horizon Readouts

First-principles electroweak closure in HQIV_LEAN

Abstract

Derived first: from the null-lattice combinatorics and horizon-area functionals, the HQIV_LEAN formalization produces a closed electroweak sector without importing PDG inputs for it—a geometric vacuum $v = T_{\text{lockin}}\, S(\texttt{referenceM} + 1)\,(1 + \gamma)$ from the lock-in temperature and outer-horizon surfaces $S(m) = (m + 1)(m + 2)$, then gauge-boson and Higgs masses from v with triality-fixed couplings; the file proves rational witnesses (e.g. $M_W^{\text{derived}} = 392/5$ in the module's units) and comparison theorems to electroweak reference values. *Carrier cross-link (aligned with "Projected complex carrier" in papers/paper/octonion_lightcone_to_oshoracle.tex):* the weak doublet is packaged in Mathlib's EuclideanSpace (Fin 8) model (Hqiv/Algebra/WeakInComplexStructure.lean); weakCarrierCinner is the explicit Hermitian slot sum and weakCarrierCinner_eq_inner rewrites it to Mathlib's inner. Hqiv/Physics/WeakDoubletCarrierGaugeQuadratic.lean defines the real Gram matrix weakWPlaneGramMatrix g v for the (σ^1, σ^2) half-Pauli directions on the VEV ray; at $(g, v) =$ (su2CouplingDerived, vacuumExpectationValueGauge it is $\lambda\,\mathbf{1}_2$, so both eigenvalues equal λ with $\mathrm{tr} = 2\lambda$ and $\det = \lambda^2$ (weakWPlaneGramMatrix_trace, weakWPlaneGramMatrix_det). The factor-8 identity M_W_derived^2 = 8 * weakWPlaneGramEigenvalue ... is then packaged with the rational witnesses boson_witness_M_W / boson_witness_m_H from Hqiv/Physics/DerivedGaugeAndLeptonSector.lean in the single certificate ew_carrier_gram_mass_certificate. A carrier quartic higgsQuarticPotentialCarrier built from higgsCarrierNormSq and the same geometric gauge vev anchor yields m_H_sq_eq_two_lambda_times_vgauge_sq with higgsQuarticLambdaGaugeWitne reconstructed from the existing mass inputs (no new tunable coupling). Triality tags So8RepIndex enter through weakPhiOfShellOnSo8Rep and weakOneOverAlphaEMAtRep; rational #eval mirrors plus example PDG-gap checks close the file. *Strong color (triplet chart):* Hqiv/Physics/QuarkColorCarrierGaugeScaffold.lean fixes the abstract 3-component chart; StrongColorSu3ChartClosure.lean exports Hermitian Gell–Mann data, half-generators T^a, and the real totally antisymmetric tensor f^{abc}; StrongColorCarrierClosure.lean conjugates 3×3 operators into the same EuclideanSpace (Fin 8) carrier as the electroweak layer. An optional Lake target lake build HQIVStrongColorSu3Certificate imports auto-generated @[simp] lemmas for nonzero f^{abc} atoms (Hqiv/Physics/StrongColorSu3fStructureSimp.lean); the default HQIVLEAN cone includes the chart files but not that certificate table. Status of the full eight-generator matrix Lie law on all pairs (a, b) is summarized in AGENTS/THEOREMS.md (strong-color table). The same backbone yields a derived neutrino ladder from $\gamma/S(\texttt{referenceM} + 2)$ applied to the neutral-vector mass. **Derived next:** a stars-and-bars lattice ratio forces $\alpha = \frac{3}{5}$ at every depth, and ratios of detuned $S(m)$ readouts drive generation steps and a rank-l^2 mass-scaling ansatz on content classes. The updated modal path now treats modal frequency / horizon data as primary through ModalFrequencyHorizon.lean, with lightweight charged-lepton ratios in LeptonResonanceGlobalDetuning.lean and outer-closure witnesses in DerivedGaugeAndLeptonSector.lean; natural indices remain evaluation coordinates. *Single-scale discipline (2026):* ScaleWitness.lean names exactly one *active* dimensionful witness per pipeline—default proton_lockin at referenceM= 4—while CODATA $1/\alpha$, CMB horizons, and PDG centrals are **predictions or comparison layers**, not simultaneous solve anchors (Section **??**). *Informational-energy closure:* InformationalEnergyMass.lean packages $E_{\text{tot}} = m_{\text{rest}} + 1/\Theta_{\text{local}}(\xi)$ with boson (additive) vs. hadron (multiplicative lapse) readout gauges; the Fano coupling mass row is $c_0 + \mathrm{loc}(\xi_G) = 2\pi\,\Omega_k(\xi_G)$ (informationalEnergyMassRow),

not the legacy vertex-share form (`informationalEnergyMassRowLegacy`). **Quarks and nucleons** use the same detuned-area ratios and lock-in binding spine; the quark *ladder* still carries explicit GeV anchors and shell tables (Section 5), but exported nucleon masses (`derivedProtonMass`, `derivedNeutronMass`) and the Python witness bundle now treat the proton lock-in readout as the primary mass/unit chart. Fermion Planck-mass labels and further SM alignment are exported in `SM_GR_Unification` and companion modules; status is tracked in `AGENTS/MASS_DERIVATION_ROADMAP.md`.

Patch QFT and discrete numerics (reader contract). HQIV is formulated as a *patch theory*: quantum fields, actions, and physical readouts are attached to discrete patches—shell indices $m \in \mathbb{N}$, finite spacetime index types (e.g. Fin 4), and horizon-limited charts—rather than to a hypothetical fundamental smooth spacetime obtained as a mesh-refinement or ultraviolet limit. Accordingly, when this text refers to "QFT," it means *patch QFT*: patching rules, finite measurement ledgers, and variational identities on the discrete spine; it is not the claim that the theory is *defined* by recovering ordinary continuum quantum field theory through such a limit. The numbers that admit the sharpest statements—mass and coupling ladders, curvature ratios, shell thermodynamics, and rational witnesses in `HQIV_LEAN`—are objects of *discrete mathematics* on $\mathbb{N}$ (combinatorial growth, cumulative simplex ledgers) together with the ordered field $\mathbb{R}$ as used in the proof assistant for analysis, bounds, and phenomenological display. Smooth-manifold or continuum field notation, when it appears, is a *translation layer* for comparison with standard physics literature; it does not supersede the discrete definitions.

1 The Core Derivation

(I) Electroweak scale from T_{lockin} and outer horizons. Let $m_{\text{lockin}} = \texttt{referenceM}$ and $T_{\text{lockin}} = T(m_{\text{lockin}}) = 1/(m_{\text{lockin}} + 1)$. `DerivedGaugeAndLeptonSector.lean` defines a geometric vacuum scale

$$v \;=\; T_{\text{lockin}} \cdot S(\texttt{referenceM} + 1) \cdot (1 + \gamma), \qquad S(m) = (m + 1)(m + 2),$$

with $\gamma = 1 - \alpha = \frac{2}{5}$. Gauge and scalar masses are built from v with triality-normalized couplings (`su2CouplingDerived`, `u1CouplingDerived`). The file proves closed rational witnesses, e.g. `boson_witness_M_W`: $M_W^{\text{derived}} = 392/5$ (GeV-normalized units in the module), and companion theorems for M_Z and m_H, together with comparison theorems to PDG centrals (`raw_ew_boson_W_H_below_PDG_Z_above`, etc.). Neutrino masses begin with `m_nu_e_derived` equal to $\big(\gamma/S(\texttt{referenceM} + 2)\big)\, M_Z^{\text{derived}}$ with cascade to higher generations (`neutrino_masses_from_outer_horizon`). *No PDG literals are imported to define these objects.*

(II) Combinatorial imprint and horizon-surface readouts. `OctonionicLightCone.lean` proves a lattice ratio equals $\alpha = \frac{3}{5}$ for every truncation (`latticeAlphaRatio_eq_alpha`). `FanoResonance.lean` packages the shared surface $S(m)$ and detuned surfaces `detunedShellSurface`; resonance steps are ratios of these, hence a concrete "$S(m)$-octave" hierarchy feeding lepton and quark ladders (`ChargedLeptonResonance`, `QuarkMetaResonance`). Where a content-class ansatz is needed, `ConservedContentMa` formulates scaling with a rank-l factor and shell correction (`massScalingAnsatz`), i.e. the expected l^2 weighting on top of the shell imprint—the combinatorial hierarchy is first-class, not an afterthought.

The current preferred lepton path is now:

$$\texttt{ModalFrequencyHorizon} \;\to\; \texttt{LeptonResonanceGlobalDetuning} \;\to\; \texttt{DerivedGaugeAndLeptonSector},$$

with `ChargedLeptonResonance` retained as a thicker phenomenology/scratch layer. This reflects the updated project stance that horizons and effective surfaces are primary, while shell numbers are sampled readouts.

(III) Variational weak/Higgs scaffold now wired to the same lock-in/curvature channel. Beyond the mass-ladder exports, `WeakHiggsFromOMaxwellScaffold.lean` now provides a typed Tier-III extension interface on the proved O-Maxwell action:

$$F^{a,\mathrm{weak}}_{\mu\nu} = (W^a_\nu - W^a_\mu) + g_w\,\mathrm{comm}^a_{\mu\nu}, \qquad \mathcal{L}_{\mathrm{weak}} = -\frac{1}{4}\sum_{a,\mu,\nu}\frac{(F^{a,\mathrm{weak}}_{\mu\nu})^2}{2},$$

with scalar potential $\lambda(|\Phi|^2 - v^2)^2$ on $\Phi \in \mathbb{R}^8$. The lock-in vev proxy is

$$v_{\mathrm{lockin}} = \sqrt{\eta_{\mathrm{paper}}\Omega_k(m_{\mathrm{lockin}}; m_{\mathrm{lockin}})}.$$

Lean now proves reduction and normalization lemmas that push this further than pure placeholders: `weakF_reduces_to_abelian`, `weakF_diag_eq_zero_of_abelian`, `lockinVev_sq_eq_eta_paper`, `lockinVev_eq_sqrt_eta_paper`, `action_O_weak_higgs_reduces_exactly_to_action_O_Maxwell`, and `weakHiggs_triality_tilt_average_eq_one`. So the status is: abelian action dynamics are proved in `Action.lean`; weak/Higgs is now a formally controlled extension surface with proved collapse back to the abelian core and proved lock-in/triality consistency identities.

2 Maxwell spectrum + Δ triality spine (what the denominator stands on)

The denominator used in the active path is no longer a free rapidity stub; it is sourced from the matrix-level O-Maxwell/Fano scaffold in `FanoOmaxwellSpectrum.lean`. The construction is:

1. **Carrier and line projector.** A Fano vertex is embedded into the octonion matrix carrier Fin 8 as index 1..7 (scalar slot 0 reserved), then lifted to a diagonal selector. Summing selectors over a line L gives

   $$\Pi_L \;=\; \sum_{v\in L}\Pi_v, \qquad \mathrm{tr}(\Pi_L) = 3$$

   (theorem `fanoLineSelector_trace`).

2. **Phase-lift generator Δ.** `PhaseLiftDelta.lean` fixes the antisymmetric 8×8 generator with nonzero $(1,7)$, $(7,1)$ entries; the scaffold proves

   $$\mathrm{tr}(\Delta\Delta^{\mathsf{T}}) = 2$$

   (theorem `phaseLiftDeltaQuadraticTrace_eq_two`).

3. **Projection normalization.** The spectral scalar is defined by

   $$\mathcal{N}(L) = \frac{\mathrm{tr}(\Pi_L) + \mathrm{tr}(\Delta\Delta^{\mathsf{T}})}{5} = \frac{3+2}{5} = 1$$

   (theorem `spectralProjectionNormalization_eq_one`).

4. **Projected mode strength and 1-jet.** For mode (L,m), projected strength is $\mathcal{N}(L)\,\texttt{phaseLiftCoeff}(m)$, and the detuning source is

$$D_L(m) \;=\; 1 + \frac{3\gamma}{2}\Big(\sigma_L(m) - \sigma_L(0)\Big).$$

Using $\mathcal{N}(L) = 1$, $\texttt{phaseLiftCoeff}(m) = \varphi(m)/6$, and $\varphi(m) = 2(m+1)$, Lean proves

$$D_L(m) = \texttt{rindlerDetuningShared}(m) = 1 + \frac{\gamma}{2}\,m$$

(`spectralFanoRindler1Jet_eq_rindler`, `spectralFanoRindler1Jet_eq_one_plus_half_gamma`).

The public denominator `trialityProjectedDenominator` in `FanoTrialityDetuningScaffold.lean` is exactly this spectral 1-jet, and `trialityProjectedDenominator_firstOrder` is therefore a proved corollary, not a placeholder.

What "triality" means here today. `Triality.lean` proves the Spin(8) representation cycle (order-3 permutation of $8v, 8s^+, 8s^-$). What is *not* yet proved is the stronger forcing theorem that the denominator's unit constant is uniquely imposed by triality-equivariant mode selection. So the current status is: *spectral Δ+Fano construction proved and affine-compatible; full triality-forcing of the constant term remains an open theorem objective.*

3 Backbone definitions (one pass)

`AuxiliaryField.lean`: $T(m) = 1/(m+1)$, $\varphi(m) = 2(m+1)$, `shell_shape`. `PhaseLiftDelta.lean`: fixed antisymmetric Δ generator and $\varphi/6$ phase-lift coefficient. `Triality.lean`: order-3 Spin(8) representation cycle bookkeeping. `SM_GR_Unification.lean`: $\alpha_{\mathrm{GUT}} = 1/42$ from cube directions $\times$ octonion imaginary units. `Baryogenesis.lean`: lock-in index `m_lockin` equals `referenceM`. `BoundStates.lean` / `HarmonicLadderMass.lean`: shell-resolved $\alpha_{\mathrm{eff}}(m)$ and hydrogenic α^2 scalings. For the variational extension path, add `Action.lean` $\to$ `WeakHiggsFromOMaxwellScaffold.lean`: proved abelian EL at base plus typed weak/Higgs layer with reduction theorems to the base action.

4 Fermion and hadron exports (what `SM_GR_Unification` names)

`SM_GR_Unification.lean` exports Planck-unit masses via `smMassFromGeometry`: scale `m_tau_Pl` divided by `resonanceProduct` (`ChargedLeptonResonance.lean`). Heavy triality slot has product 1, so τ, top, and bottom labels coincide at that functional's value.

5 Single-scale witness and informational-energy mass row

One active witness per pipeline. `Hqiv/Physics/ScaleWitness.lean` defines an inductive `ScaleWitness` with three modes: `proton_lockin` (default), `codata_alpha`, and `cmb_now`. The discipline is operational, not a new Mathlib axiom: in any given export or Python solve, **exactly one** of these may pin the dimensionful chart; the others are cross-checks.

- `proton_lockin` — `derivedProtonMass` at `referenceM` fixes the mass/unit chart; neutron mass follows from the same lock-in shell and constituent content (`derivedDeltaM` from udd vs. uud); W/Z/H masses load from the Lean closure witnesses in `data/hqiv_witnesses.json`. CODATA $1/\alpha$ is a *predicted* EM coupling, not a row in the solve.

- `codata_alpha` — legacy regression mode: the continuous Gauss→EW brace pins $1/\alpha$ to CODATA while geometry supplies the rest (Python `--scale-witness codata_alpha` with continuous ξ brace).

- `cmb_now` — shallow-ξ comparison chart ($\xi \approx 1.07$ mean-field); not the brace mass row at lock-in ($\xi = 5$).

Export: `lake build HQIVWitnesses`; metadata patch `scripts/export_witnesses_metadata.lean`; Python loader `scripts/hqiv_scale_witness.py`.

Informational energy and readout gauges. In natural units ($c = \hbar = T_{\rm Pl} = 1$), `InformationalEnergyMass.l` defines

$$E_{\rm tot}(m_{\rm rest}, \xi) = m_{\rm rest} + \frac{1}{\Theta_{\rm local}(\xi)}, \qquad \Theta_{\rm local}(\xi) = \frac{T_{\rm Pl}}{\xi}, \qquad \mathrm{loc}(\xi) = \frac{1}{\Theta_{\rm local}(\xi)} = \xi.$$

Bosons use `additiveLocalization` (full $E_{\rm tot}$ is the observable mass); hadrons use `multiplicativeLapse` / `hadronMassFromXi` (rest slot $\div N_{\rm lapse}$ with localization carried by the lapse increment, not double-counted). `gauge_transformation_localization_to_lapse` proves when the two gauges agree after calibrating $m_{\rm rest}$.

Mass row on the Fano coupling system. `ContinuousXiCoupling.lean` packages the Python `hqiv_coupling_linear_system.py` rows in Lean form. The **default** mass-sector row is

$$c_0 + \mathrm{loc}(\xi_G) = 2\pi\,\Omega_k(\xi_G), \qquad \Omega_k(\xi) = \frac{I(\xi)}{I(\xi_{\rm lock})},$$

with $\xi_{\rm lock} =$ `referenceM`$+1 = 5$ so $\Omega_k(\xi_{\rm lock}) = 1$. Theorem `informationalEnergyMassRow_budget` states the budget identity; when the row holds, `informationalEnergy_satisfied_when_row_holds` gives $E_{\rm tot} = 2\pi\,\Omega_k(\xi_G)$ and `informationalEnergy_over_twoPi_eq_omegaK_when_row_holds` gives $E_{\rm tot}/(2\pi) = \Omega_k$. The older form `informationalEnergyMassRowLegacy` ($c_0$+loc = `holonomyRowRhs`$(0)\cdot \Omega_k$) is retained for regression only.

Three Ω_k charts (do not conflate). Agents must keep separate: (i) **brace/coupling** $\Omega_k(\xi_G) \approx 0.72$ at $\xi_G \approx 3.47$ with $\xi_{\rm lock} = 5$; (ii) **shallow** $\Omega_k(\xi \approx 1.07) \approx 0.03$ (not the brace root); (iii) $\Omega_k^{\rm true}$ axiom limit ≈ 0.0098 and CMB-as-horizon scales $\sim 10^{-28}$—comparison layers, not slots in the brace mass row.

Python coherence checks. Under `proton_lockin`, the coupling solve normalizes $c_0 = 1$ and reports braced $1/\alpha$ as a prediction vs. CODATA; under `codata_alpha`, CODATA pins the brace. `scripts/informational_energy_mass.py` attaches particle rows (proton, neutron, W, Z, H) at solved ξ_v with masses from the witness JSON, not scattered literals.

6 Quarks and nucleons: what is derived, what is still a witness

Honesty (not "no witnesses"). The quark ladder in `QuarkMetaResonance.lean` is *not* free of external numerals. In particular:

- **Absolute scale:** the up-type ladder is normalized by a named real `m_top_GeV` (PDG-style pole-mass listing used as calibration witness). Ratios between generations are *not* taken from PDG tables; they are fixed by `geometricResonanceStep` ratios built from detuned surfaces `detunedShellSurface` and explicit $\mathbb{N}$ shell endpoints (`m_quark_up_*`).
- **Readout tables:** the integer triples `m_quark_up_*` and `m_quark_down_*` are calibration inputs that determine internal resonance factors; uniqueness of these coordinates from the null lattice alone is *not* proved in this module.
- **Legacy down branch:** a separate real `m_bottom_GeV` still fixes the old down-type comparison ladder (`quarkMassDown`, `m_strange_GeV`, `m_down_GeV`) unless/until that path is replaced end-to-end by the active API.

Lock-in readout, `T_lockin`, and the GeV top anchor (disambiguation). The lock-in indices agree across modules: `m_lockin` = `referenceM` = `m_top_at_lockin` (`Baryogenesis`, `QuarkMetaResonance`), and

$$\texttt{T\_lockin} = T(\texttt{m\_top\_at\_lockin})$$

on the auxiliary ladder (`AuxiliaryField.lean`: $T(m) = T_{\mathrm{Pl}}/(m+1)$ with $T_{\mathrm{Pl}} = 1$ in natural units). `SM_GR_Unification` records this as `T_lockin_eq_temperature_at_quark_top_birth_shell`. So the **story** is exactly what you describe: lock-in, baryogenesis, and the "top birth" readout are the *same* calibration row, and `T_lockin` is the horizon temperature assigned to that row. That is *not* the same proposition as "the real number `T_lockin` equals the pole mass m_t in GeV": at `referenceM`= 4 one has $T = 1/5$ in those units, not ~ 172 GeV. The explicit real `m_top_GeV` exists precisely to supply **laboratory GeV** normalization for the color-resonance ladder (`topLockinColorResonanceAnchorMass`, etc.); it is the witness that remains separate from the abstract ladder unless a future theorem closes the gap. A parallel export `m_top_quark_natural` from `smMassFromGeometryLabel .top` uses the τ Planck mass and resonance products only—no import of `m_top_GeV` there. ADM lapse and plasma corrections (`HQVM_lapse`; `QuarkMetaResonance` docstring) are *not* applied inside `T` itself, so "top energy" in the physical paper sense may require that layer; the Lean `T_lockin` identity is the lock-in temperature at that readout.

Why top is the heavy anchor (current implementation). The active color-composed API fixes the heavy up-like channel by `allowedColorResonanceMass .upLike .heavy = m_top_GeV`. This is not arbitrary bookkeeping: it matches the lock-in-heavy channel where the resonance product is 1, so the heavy band is the unique non-relaxed reference from which mid/light bands are generated by the same detuned-surface readout map. In other words, top is used as the anchor because the current ladder is implemented as *"heavy lock-in witness + horizon/surface relaxations"* rather than as three independently normalized flavor masses.

What *is* derived in-structure from the same spine as leptons ($S(m)$, Rindler detuning, γ, α, `referenceM`) are the **ratio factors** between shells and the **composite-trace** binding `nucleonSharedBinding_MeV` at `referenceM`.

Active visible-state API (color-composed ladder). The public masses `allowedColorResonanceMass` normalize the heavy up-like band to the top anchor, then compress the heavy down-like visible mass by a cross-channel detuning

$$\texttt{crossChannelHeavyShellDetuning} = \frac{\texttt{detunedShellSurface}(\texttt{m\_quark\_up\_top\_shell})}{\texttt{detunedShellSurface}(\texttt{m\_quark\_down\_bottom\_shell})}$$

and a visible bookkeeping factor `chargedLeptonContentCount`$\times$`cubeAxes` $= 2 \times 3$ (two visible charge signs across three color channels). Mid/light bands descend by the same `downResonanceProduct` / `upResonanceProduct` machinery as elsewhere. Nucleon constituents use the active light masses `activeUpLightQuarkMass_GeV`, `activeDownLightQuarkMass_GeV` with a shared dressing scale and a smaller residual detuning tied to `alphaEffAtShell referenceM`; `DerivedNucleonMass.lean` exports `derivedProtonMass`, `derivedNeutronMass`, and `derivedDeltaM`. Under `proton_lockin` (Section ??), the exported proton mass is the **primary** mass witness; `protonAnchorMass_MeV`= 938.272 in `QuarkMetaResonance.lean` remains a legacy reference row for lapse readout lemmas (`LapseMassReadout.lean`), not a second simultaneous anchor in the Python pipeline. The neutron–proton gap is content-only at shared lock-in ξ: udd vs. uud constituent energies with shared `nucleonSharedBinding_MeV` cancelling in Δm.

Representative numerics (module units, current witness set). Evaluating the definitions with the literals in the repository (same formulas as `QuarkMetaResonance.lean`; approximate) gives:

charm (from top / first up-step)	≈ 1.27 GeV
active up-like light (`allowedColorResonanceMass upLike light`)	≈ 0.00220 GeV
active down-like heavy (compressed)	≈ 4.88 GeV
active down-like light	≈ 0.00550 GeV
`derivedProtonMass`	≈ 938.3 MeV
`derivedNeutronMass`	≈ 939.6 MeV
`derivedDeltaM`	≈ 1.29 MeV

These are **not** PDG matches; they are the output of the closed Lean definitions given the current anchors and readout tables.

Why they differ from observation. PDG values for quarks and hadrons are quoted in different schemes (pole vs. $\overline{\text{MS}}$, running scales) and include nonperturbative QCD and finite-volume/lattice input where applicable. This formalization instead implements a **finite-chart resonance surrogate**: fixed $\mathbb{N}$ readout endpoints, a single heavy-quark GeV anchor, and a composite-trace binding ansatz at `referenceM`. Differences are therefore expected:

- **Proton $\sim$ 938 MeV:** the current bottom-up network export lands near the laboratory anchor when the constituent/binding split and active light-quark readouts are applied at `referenceM`; this is still a finite-chart surrogate, not a lattice QCD calculation. Residual scheme dependence (pole vs. $\overline{\text{MS}}$, running scale) remains outside the Lean artifact.

- **Light quark "masses":** experimental light-quark masses are indirect and scheme-dependent; our GeV numbers are internal ladder coordinates, not a claim of pole masses for u, d.

- **Heavy flavors:** charm/bottom/top comparisons would require either removing the top anchor or deriving it from a deeper lock-in theorem; until then, agreement with PDG is not the target of the proof artifact.

7 Explicit inputs and proof obligations (AGENTS-aligned)

The file `AGENTS/MASS_DERIVATION_ROADMAP.md` is the authoritative status board. In short:

- **Faithful to the light-cone spine today:** bosonic closure witnesses and neutrino witnesses in `DerivedGaugeAndLeptonSector.lean`; structural binding quantities (`nucleonSharedBinding_MeV`, composite-trace packaging) tied to `referenceM`, α, γ, and $S(m)$; informational-energy readout gauges and the $2\pi\,\Omega_k$ mass row in `InformationalEnergyMass.lean` / `ContinuousXiCoupling.lean` (Section **??**).
- **Single-scale export discipline:** `ScaleWitness.lean` + `data/hqiv_witnesses.json` + `scripts/hqiv_scale_witness.py`; default active witness `proton_lockin`; CODATA α and PDG masses are comparison targets unless `codata_alpha` mode is explicitly selected.
- **Variational extension now formalized as scaffold:** `WeakHiggsFromOMaxwellScaffold.lean` defines non-abelian-form weak field strength, Higgs potential, lock-in vev proxy, and combined action slot, with proved reduction identities back to `action_O_Maxwell_general`.
- **Still carried by explicit numerals or boundary conditions:** substrate pins (`qcdShell`, `latticeStepCount` ⇒ `referenceM`); **quark ladder:** `m_top_GeV`, legacy `m_bottom_GeV`, and the $\mathbb{N}$ shell triples in `QuarkMetaResonance.lean` (Section 5); several SM alignment decimals in `SM_GR_Unification.lean` where the code matches paper fiducials by definition.
- **What the roadmap asks next:** reduce *quark* anchor count and unify the legacy down ladder with the active visible-state API; derive or eliminate `m_top_GeV` without breaking the color-composed API; for the variational side, promote the weak/Higgs scaffold from reduction-level theorems to full EL closure on weak/scalar channels—see `AGENTS/MASS_DERIVATION_ROADMAP.md`.

This division is intentional: the formalization *separates* closure witnesses derived from $(T_{\text{lockin}}, S, \gamma, \alpha)$ from numerals whose job is alignment or unfinished closure. Readers evaluating the program should use the roadmap's honesty table, not footnotes in a survey note.

8 Octonions and the Standard Model bookkeeping layer

`SMEmbedding.lean`, `Triality.lean`, `Generators.lean`: octonionic 8-dimensional carriers, $\mathfrak{so}(8)$ generators, and Standard Model quantum numbers. This layer fixes representation labels; the dynamical masses above come from the shell program in Sections 1–5. In parallel, the active color triplet is packaged on the 3×3 chart (`QuarkColorCarrierGaugeScaffold.lean`, `StrongColorSu3ChartClosure.le` and embedded into the octonion-indexed carrier via `StrongColorCarrierClosure.lean`; optional `HQIVStrongColorSu3Certificate` supplies generated simp lemmas for nonzero f^{abc} without enlarging the default `HQIVLEAN` glob.

Reading order in the repository

`OctonionicLightCone.lean` → `AuxiliaryField.lean` → `PhaseLiftDelta.lean` → `FanoLine.lean` → `OMaxwellAlgebraSeed.lean` → `FanoOmaxwellSpectrum.lean` → `FanoTrialityDetuningScaffold.lean` → `ContinuousXiCoupling.lean` → `InformationalEnergyMass.lean` → `ScaleWitness.lean` → `ModalFrequencyHorizon.lean` → `LeptonResonanceGlobalDetuning.lean` → `DerivedGaugeAndLeptonSector.l` → `FanoResonance.lean` / `QuarkMetaResonance.lean` (Section 5) → `DerivedNucleonMass.lean` → `LapseMassReadout.lean` → `SM_GR_Unification.lean`; Python witness/coherence: `scripts/hqiv_scale_witne` `scripts/hqiv_coupling_linear_system.py`, `scripts/informational_energy_mass.py`; color triplet chart + carrier embed: `QuarkColorCarrierGaugeScaffold.lean` → `StrongColorSu3ChartClosure.lean` → `StrongColorCarrierClosure.lean` (optional cert: `lake build HQIVStrongColorSu3Certificate`);

variational branch: `Action.lean` $\to$ `WeakHiggsFromOMaxwellScaffold.lean`; optional thicker lepton phenomenology: `ChargedLeptonResonance.lean`; heavy $\mathfrak{so}(8)$ matrix closure: `lake build HQIVSO8Closure`; parallel: `AGENTS/MASS_DERIVATION_ROADMAP.md`, `AGENTS/THEOREMS.md`, `AGENTS/ASSUMPTIONS.m`

Acknowledgments

Mathlib supplies analysis and linear algebra. Maintainership notes: `AGENTS/ASSUMPTIONS.md`, `AGENTS/THEOREMS.md`.